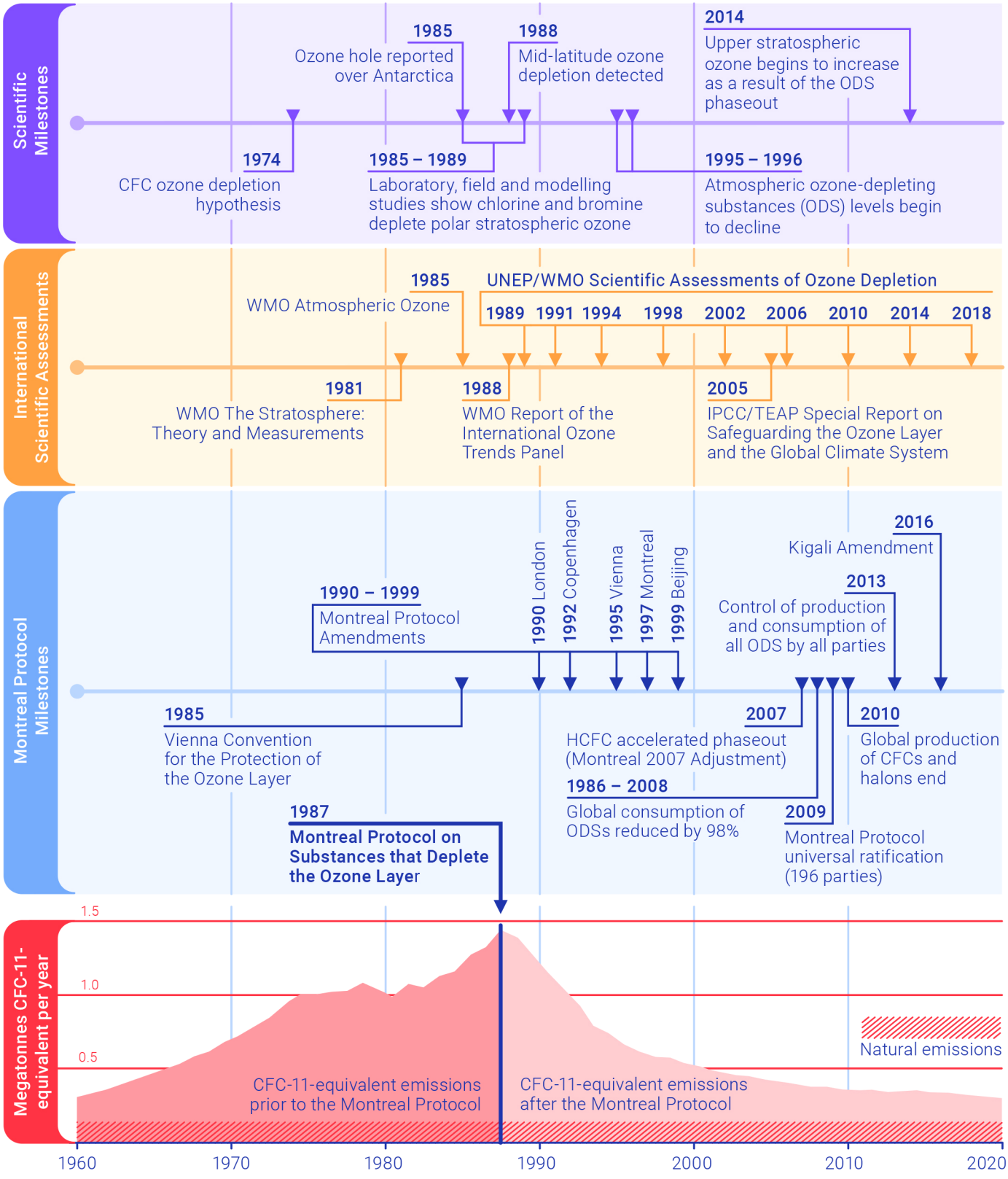
Scientific Milestones
1974
CFC ozone depletion hypothesis
1985
Ozone hole reported over Antarctica
1985 – 1989
Laboratory, field and modelling studies show chlorine and bromine deplete polar stratospheric ozone
1988
Mid-latitude ozone depletion detected
1995 – 1996
Atmospheric ozone-depleting substances (ODS) levels begin to decline
2014
Upper stratospheric ozone begins to increase as a result of the ODS phaseout
International Scientific Assessments
1981
WMO The Stratosphere: Theory and Measurements
1985
WMO Atmospheric Ozone
1988
WMO Report of the International Ozone Trends Panel
UNEP/WMO Scientific Assessments of Ozone Depletion
1989 1991 1994 1998 2002 2006 2010 2014 2018
2005
IPCC/TEAP Special Report on Safeguarding the Ozone Layer and the Global Climate System
Montreal Protocol Milestones
1985
Vienna Convention for the Protection of the Ozone Layer
1987
Montreal Protocol on Substances that Deplete the Ozone Layer
1990 – 1999
Montreal Protocol Amendments
1990 London
1992 Copenhagen
1995 Vienna
1997 Montreal
1999 Beijing
1986 – 2008
Global consumption of ODSs reduced by 98%
2007
HCFC accelerated phaseout (Montreal 2007 Adjustment)
2009
Montreal Protocol universal ratification (196 parties)
2010
Global production of CFCs and halons end
2013
Control of production and consumption of all ODS by all parties
2016
Kigali Amendment
Megatonnes CFC-11-equivalent per year
1.5
1.0
0.5
CFC-11-equivalent emissions prior to the Montreal Protocol
CFC-11-equivalent emissions after the Montreal Protocol
Natural emissions
1960
1970
1980
1990
2000
2010
2020